

Abhishri Ajit Medewar

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EDUCATION

Master of Computer Science, Arizona State University

August 2021 - May 2023

Relevant Coursework: Machine Learning, Deep Learning, Cloud Computing

GPA 3.8

Bachelor of Computer Engineering, Pune University

August 2014 - May 2018

Relevant Coursework: Data Structures, Database Management Systems, Software Engineering

GPA 3.7

SKILLS

Programming Languages: Python, C++, C

Frameworks and Libraries: PyTorch, Tensorflow, Scikit-learn, Numpy, Scipy, Pandas, Matplotlib, Seaborn, Keras, OpenCV, Pillow, Albumentations, BeautifulSoup, AWS, Boto3, Docker, S3, Hadoop, Kubernetes, MLflow, Weights & Biases, ONNX, TensorRT, HuggingFace, LlamaIndex, LangChain, Ollama, Chroma DB, Faiss

LLM - Parameter-Efficient Fine-Tuning (PEFT), Low-Rank Adaptation (LoRA), Retrieval-Augmented Generation (RAG)

Web Technology and Database: Flask, FastAPI, React, Angular, HTML, CSS, MySQL, PostgreSQL, MongoDB

Operating System/Developer Tools: Linux, Windows, Git, Jira, VS Code, Sublime text, Visual Studio, PyCharm

EXPERIENCE

Machine Learning Engineer, vivoVerse, Austin, USA

October 2023 - Present

- Implemented **LoRA for parameter-efficient fine-tuning** of the ViT-Huge classification model, using only 1% of the trainable parameters and achieving **90% accuracy with 400 training samples**.
- Implemented a **2.5D U-Net** architecture, achieving **98% accuracy** in segmenting **6000x400 images in 0.2 seconds**, reducing manual effort by **1500x**.
- Developed and deployed an object tracking model, improving **object detection accuracy by 2%**. Converted the **model to ONNX format** to enhance inference speed by 30% and facilitate seamless deployment in a production environment.
- Implemented and integrated **data pipelines into ML platform** to facilitate model training for large scale (terabytes) datasets. Utilized **MLflow** for comprehensive testing, version control, and real-time monitoring of machine learning models.

Machine Learning Engineer Intern, Intel Corporation, Santa Clara, USA

June 2022 - April 2023

- Implemented a **vision transformer** from scratch for classification of **images with dimensions of 5000x3000**, leading to a **5% enhancement** in classification accuracy.
- Implemented **distributed training** and hyperparameter optimization for a video frame interpolation model, achieving a **20% reduction in training time** and a 2% performance improvement.

Machine Learning Engineer, Tata Elxsi, Pune, India

July 2018 - June 2021

- Implemented State-of-the-Art region-based object detector using **Faster-RCNN** with **98% accuracy** and average inference time of 3 second/image on **Jetson TX2**, with image dimension of 2592x1944.
- Deployed an InceptionV2 and Faster-RCNN based object detector on **Nvidia Jetson TX2** using Tensorflow C++ API and bazel.
- Implemented **K-means clustering algorithm** for detecting small dust particles on camera lens, anti-reflection glass for optical system with overall accuracy of **95%**.

PROJECTS

Retrieval-Augmented Generation App Using LangChain and Llama Model [[Github](#)]

Fall 2024

- Developed a **RAG** application enabling users to upload research paper PDFs and ask questions about the content.
- Utilized the open source **Llama 3.1, Ollama, Langchain** and **Chroma DB** for precise content retrieval and understanding.

AI Based Image Background Segmentation and Editing [[Github](#)]

Spring 2023

- Implemented **U-Net** architecture from scratch, achieving **98.90% accuracy (dice score)** in background segmentation using the Human Segmentation Dataset.
- Employed **TensorRT** to convert the trained model into an optimized TensorRT format, resulting in 6x performance boost.

Scalable Application Deployment with Kubernetes and Docker [[Github](#)]

Fall 2022

- Developed an ML application for handwritten digits classification, using Flask server and Docker for seamless deployment.
- Utilized **Kubernetes** clusters on **AWS EKS** and **GCP GKE** for streamlined deployment and performance monitoring.

Deep Learning Approach for Human Stress Detection [[Github](#)]

Summer 2018

- Developed a custom **CNN** for detection of stress level (0-9) by using FER2013 dataset to **classify** the facial emotion of the person from an image and correlating the emotion percentage with stress level **with 85% accuracy**.
- Utilized **Flask** to develop **APIs** for serving models, facilitating seamless deployment of model.

ACHIEVEMENTS

- Received **Tata Elxsi Bravo Award** for outstanding work in machine learning model fine-tuning.
- Exhibited strong leadership by driving **Proof of Concept** work at Tata Elxsi and **securing funding** from key clients.